

Module 1: Introduction to Penetration Testing



Topic 2: Introduction to Kali Linux



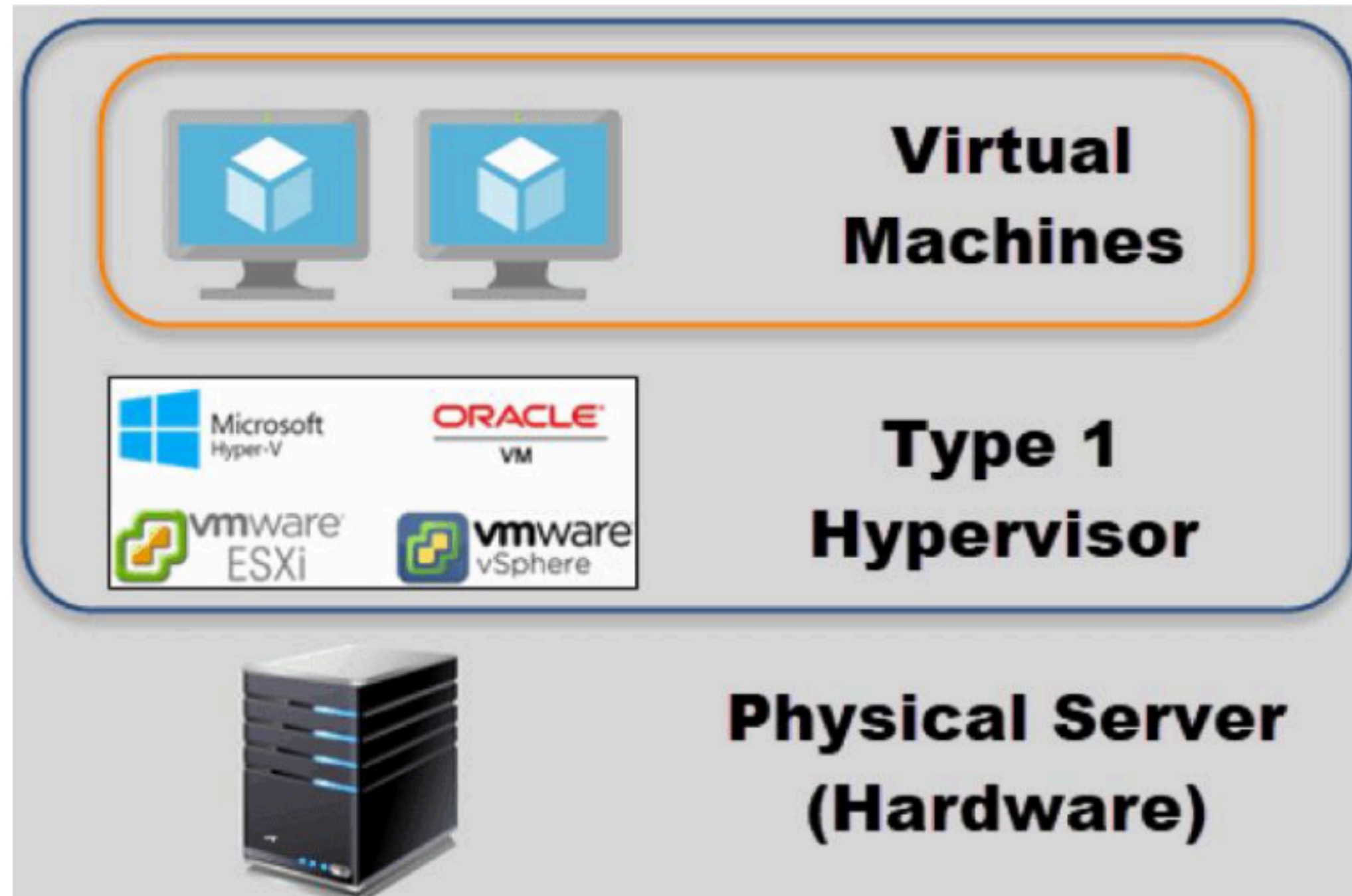
What is Kali Linux?

It is a Debian-based operating system that comes pre-loaded with a wide range of tools and software for various aspects of cybersecurity, including network analysis, vulnerability assessment, digital forensics, and ethical hacking.

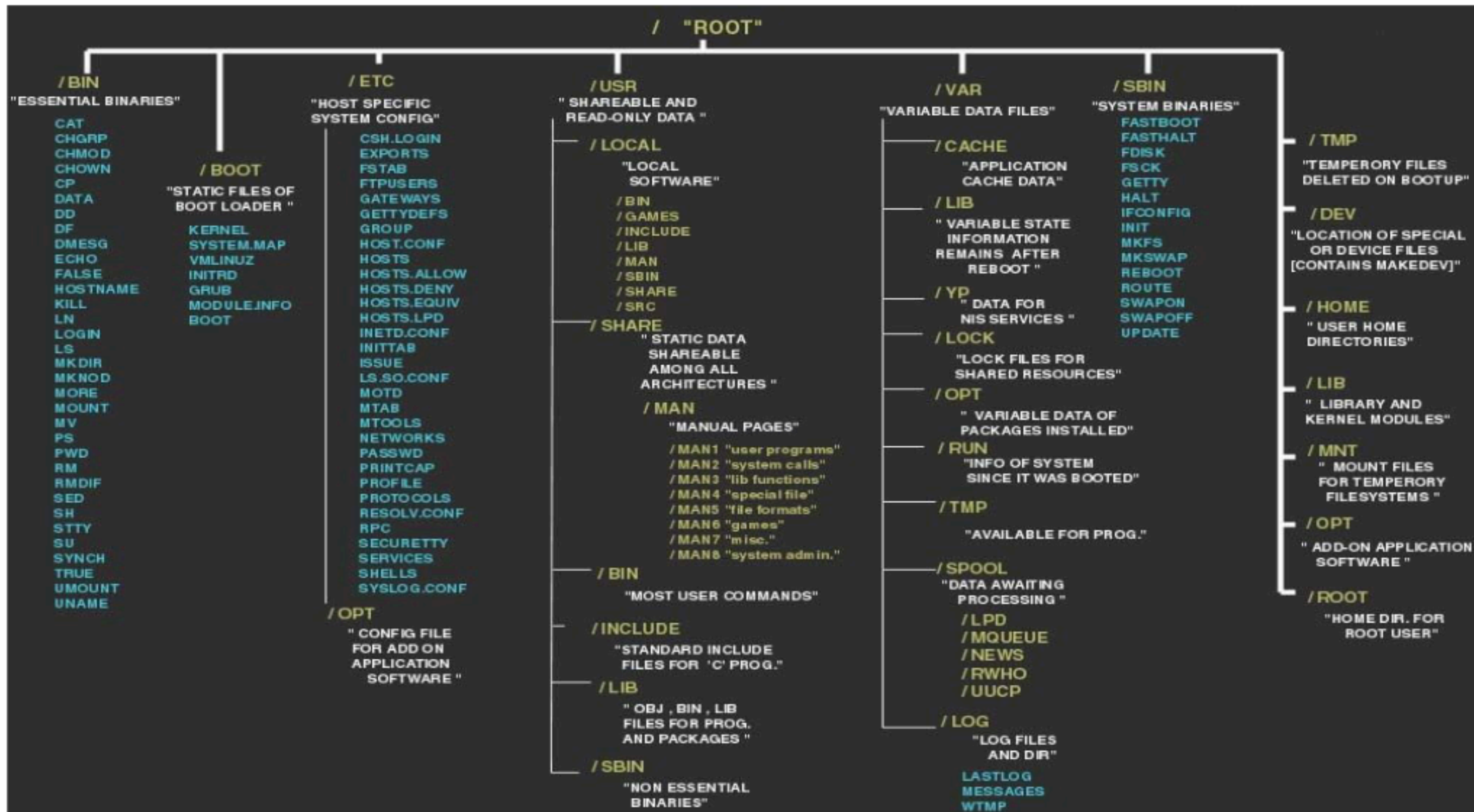
Key features and components of Kali Linux include:

- Security Tools: Kali Linux includes over 600 pre-installed security tools
- Customization: Kali Linux allows users to customize their installations and choose which tools to include based on their specific needs.
- Live Boot: Kali Linux can be run as a live bootable system from a USB drive or DVD, enabling users to use it on different computers without installation.

Getting Linux For Your Computer



Linux File System (Filesystem Hierarchy Standard (FHS))



Linux Command Line

Basically, a command is an input typed at the shell to perform a specific task.

The general form of Linux commands:

```
command -option(s)[optional] argument(s)[optional]
```

Notes:

- Linux commands are case-sensitive.
- Options and arguments are optional, and that means that not all the commands require them.

Navigating the File System

`pwd` print working directory
`cd` change directory

```
root@nitro:/home/condor| => pwd
/home/condor
root@nitro:/home/condor| => cd ..
root@nitro:/home| => pwd
/home
root@nitro:/home| => cd ..
root@nitro:/| => pwd
/
root@nitro:/| => cd home
root@nitro:/home| => cd condor
root@nitro:/home/condor| => pwd
/home/condor
root@nitro:/home/condor| => 
```


Navigating the File System

- ls list the files
- ls -a hidden files
- ls -l long listing

```
root@nitro:/home/condor/pentest| => ls
test1 test2
root@nitro:/home/condor/pentest| => ls -a
. .. .secret test1 test2
root@nitro:/home/condor/pentest| => ls -l
total 8
drwxr-xr-x 2 root root 4096 Oct 13 15:32 test1
drwxr-xr-x 2 root root 4096 Oct 13 15:32 test2
root@nitro:/home/condor/pentest| => ls -al
total 52
drwxr-xr-x  5 root    root    4096 Oct 13 15:33 .
drwxr-xr-x 49 condor  condor 36864 Oct 13 15:27 ..
drwxr-xr-x  2 root    root    4096 Oct 13 15:33 .secret
drwxr-xr-x  2 root    root    4096 Oct 13 15:32 test1
drwxr-xr-x  2 root    root    4096 Oct 13 15:32 test2
root@nitro:/home/condor/pentest| => 
```


Creating and Removing Files/Directories

mkdir make directory

rmdir remove directory

```
root@nitro:/home/condor => mkdir pentest
root@nitro:/home/condor => mkdir pentest/cybertalents
root@nitro:/home/condor => ls pentest
cybertalents
root@nitro:/home/condor => rmdir pentest/cybertalents
root@nitro:/home/condor => ls pentest
root@nitro:/home/condor => 
```

Users and Groups

Command	Description
id	It is used to find out user and group names and numeric IDs (UID or GID) of the current user or any other user.
whoami	It displays the username of the current user when this command is executed.
adduser	Add a user to the system.
addgroup	Add a group to the system.
userdel	Delete a user account and related files.
groupdel	Delete a group.
passwd	Change user password.

Permissions

```
root@nitro:/home/condor/pentest|⇒ ls -al /tmp
total 116
drwxrwxrwt 25 root root 12288 Oct 13 15:47 .
drwxr-xr-x 23 root root 4096 Sep  5 23:19 ..
drwx----- 2 condor condor 4096 Oct 13 12:59 .com.google.Chrome.N06YzI
drwx----- 2 condor condor 4096 Oct 13 12:40 'com.microsoft.teams.linux Crashes'
-rw----- 1 condor condor 0 Oct 13 12:40 config-err-hGPRyI
-rw----- 1 condor condor 535 Oct 13 12:40 dropbox-antifreeze-PU0iH0
drwxrwxrwt 2 root root 4096 Oct 13 12:40 .font-unix
drwxrwxrwt 2 root root 4096 Oct 13 12:40 .ICE-unix
```


Batch Scripting

```
root@nitro:/home/condor/pentest| => ./sweep.sh
192.168.0.1
192.168.0.105
192.168.0.142
192.168.0.106
192.168.0.173
192.168.0.115
192.168.0.175
```